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1
2 clc;
3 clear;
4 close all;
5
6 %% ===== P10-3: TBA Production Rate Laws =====
7 fprintf('===== P10-3: TBA Production Rate Laws =====\n\n');
8
9 % Part (a): Surface reaction is rate-limiting
10 fprintf('(a) Surface reaction rate-limiting:\n');
11 fprintf('    Rate law:  $r_{TBA} = k \cdot K_I \cdot K_W \cdot C_I \cdot C_W / (1 + K_I \cdot C_I + K_W \cdot C_W)^2$ \n\n');
12
13 % Part (b): Adsorption of isobutene is limiting
14 fprintf('(b) Isobutene adsorption limiting:\n');
15 fprintf('    Rate law:  $r_{TBA} = k \cdot C_I \cdot C_S / (1 + K_W \cdot C_W + K_{TBA} \cdot C_{TBA})$ \n\n');
16
17 % Part (c): Eley-Rideal kinetics
18 fprintf('(c) Eley-Rideal kinetics:\n');
19 fprintf('    Rate law:  $r_{TBA} = k \cdot K_I \cdot C_I \cdot C_W / (1 + K_I \cdot C_I)$ \n\n');
20
21 % Part (d): Different adsorption sites
22 fprintf('(d) Different adsorption sites:\n');
23 fprintf('    Rate law:  $r_{TBA} = k \cdot [C_I \cdot C_W - C_{TBA} / K_s] / [(1 + K_W \cdot C_W)(1 + K_I \cdot C_I)]$ \n\n');
24
25 % Part (e): Generalization
26 fprintf('(e) Generalization:\n');
27 fprintf('    All rate laws have the form:\n');
28 fprintf('     $r_{TBA} = (\text{numerator with reactants}) / (\text{denominator with adsorption terms})$ \n');
29 fprintf('    - Denominator power depends on number of sites in rate-limiting step\n');
30 fprintf('    - Eley-Rideal has simpler form (only one adsorbed species)\n\n');
31
32 %% ===== P10-10: MEK Production =====
33 fprintf('\n===== P10-10: MEK Production from Butanol =====\n\n');
34

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%% ===== P10-10: MEK Production =====

fprintf('\n===== P10-10: MEK Production from Butanol =====\n\n');

% Given experimental data

P_Bu = [2, 0.1, 0.5, 1, 2, 1]; % atm

P_MEK = [5, 0, 2, 1, 0, 0]; % atm

P_H2 = [0, 0, 1, 1, 0, 10]; % atm

r_MEK = [0.044, 0.040, 0.069, 0.060, 0.043, 0.059]; % mol/h/g-cat

% Part (a): Suggest a rate law

fprintf('(a) Rate Law Analysis:\n');

rate_per_PBu = r_MEK ./ P_Bu;

fprintf(' r_MEK/P_Bu values: ');

fprintf('%0.3f ', rate_per_PBu);

fprintf('\n Average = %0.3f\n', mean(rate_per_PBu));

fprintf(' Suggested rate law: $r_{\text{MEK}} = k * P_{\text{Bu}}$ \n\n');

% Part (b): Reaction mechanism

fprintf('(b) Reaction Mechanism:\n');

fprintf(' 1. Bu + S <-> Bu*S (adsorption, equilibrium)\n');

fprintf(' 2. Bu*S -> MEK*S + H2*S (surface reaction, RATE-LIMITING)\n');

fprintf(' 3. MEK*S <-> MEK + S (desorption, fast)\n');

fprintf(' 4. H2*S <-> H2 + S (desorption, fast)\n\n');

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% Part (c): Six ideas

fprintf('(c) Six Ideas from Preface Table P-4:\n');

fprintf(' 1. Use systematic algorithm for rate law derivation\n');

% Part (d): Calculate conversion vs catalyst weight

fprintf('(d) Conversion Calculation:\n');

F_Bu0 = 10; % mol/min

P0 = 10; % atm

W_max = 23; % kg

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% Find W for 90% conversion
W_90 = -F_Bu0 / (k_apparent * P0) * log(1 - 0.90);
fprintf('    Catalyst weight for 90%% conversion: W = %.2f kg\n\n', W_90);

% Plot
figure('Position', [100, 100, 1000, 400]);
subplot(1, 2, 1);
plot(W, X, 'b-', 'LineWidth', 2);
hold on;
plot(W_90, 0.90, 'ro', 'MarkerSize', 10, 'LineWidth', 2);
xlabel('Catalyst Weight (kg)');
ylabel('Conversion, X');
title('Conversion vs Catalyst Weight');
grid on;
legend('Conversion', '90% Target');

subplot(1, 2, 2);
r_reactor = k_apparent * P0 * (1 - X);
plot(W, r_reactor, 'r-', 'LineWidth', 2);
xlabel('Catalyst Weight (kg)');
ylabel('Reaction Rate (mol/min/kg)');
title('Reaction Rate vs Catalyst Weight');
grid on;

% Part (e): Critical thinking
fprintf('(e) Critical Thinking Question:\n');
fprintf('    Question: Is our rate law valid at all pressures?\n\n');
fprintf('    Why this requires critical thinking:\n');
fprintf('    - We assume linear dependence on P_Bu (weak adsorption)\n');
fprintf('    - Data only spans P_Bu = 0.1 to 2 atm\n');
fprintf('    - At high P_Bu, saturation effects may occur\n');
fprintf('    - At low P_Bu, mass transfer may limit the rate\n');
fprintf('    - Extrapolation beyond data range is risky\n');

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alpha = 0.03; % kg^-1
eps = 1; % 1 mol Bu -> 2 mol products (MEK + H2)

% ODE system: dX/dW and dy/dW where y = P/P0
odefun = @(W, Y) [
    k_apparent * P0 * Y(2) * (1 - Y(1)) / F_Bu0; % dX/dW
    -alpha / (2 * Y(2)) * (1 + eps * Y(1)) % dy/dW
];

[W_sol, Y_sol] = ode45(odefun, [0, W_max], [0, 1]);
X_sol = Y_sol(:, 1);
y_sol = Y_sol(:, 2);
P_sol = y_sol * P0;

% Find W for 90% conversion
idx = find(X_sol >= 0.90, 1);
if ~isempty(idx)
    fprintf(' W for 90%% conversion = %.2f kg\n', W_sol(idx));
    fprintf(' Pressure at 90%% conversion = %.2f atm\n\n', P_sol(idx));
end

% Plot results
figure('Position', [100, 100, 1200, 400]);

subplot(1, 3, 1);
plot(W_sol, X_sol, 'b-', 'LineWidth', 2);
hold on;
plot(W, X, 'r--', 'LineWidth', 1.5);
xlabel('Catalyst Weight (kg)');
ylabel('Conversion, X');
title('Conversion Profile');
legend('With pressure drop', 'Without pressure drop');
grid on;

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===== P10-3: TBA Production Rate Laws =====

(a) Surface reaction rate-limiting:

$$\text{Rate law: } r_{\text{TBA}} = k \cdot K_I \cdot K_W \cdot C_I \cdot C_W / (1 + K_I \cdot C_I + K_W \cdot C_W)^2$$

(b) Isobutene adsorption limiting:

$$\text{Rate law: } r_{\text{TBA}} = k \cdot C_I \cdot C_S / (1 + K_W \cdot C_W + K_{\text{TBA}} \cdot C_{\text{TBA}})$$

(c) Eley-Rideal kinetics:

$$\text{Rate law: } r_{\text{TBA}} = k \cdot K_I \cdot C_I \cdot C_W / (1 + K_I \cdot C_I)$$

(d) Different adsorption sites:

$$\text{Rate law: } r_{\text{TBA}} = k \cdot [C_I \cdot C_W - C_{\text{TBA}}/K_s] / [(1 + K_W \cdot C_W)(1 + K_I \cdot C_I)]$$

(e) Generalization:

All rate laws have the form:

$$r_{\text{TBA}} = (\text{numerator with reactants}) / (\text{denominator with adsorption terms})$$

- Denominator power depends on number of sites in rate-limiting step
- Eley-Rideal has simpler form (only one adsorbed species)

===== P10-10: MEK Production from Butanol =====

(a) Rate Law Analysis:

$r_{\text{MEK}}/P_{\text{Bu}}$ values: 0.022 0.400 0.138 0.060 0.021 0.059

Average = 0.117

Suggested rate law: $r_{\text{MEK}} = k \cdot P_{\text{Bu}}$

(b) Reaction Mechanism:

1. $\text{Bu} + \text{S} \rightleftharpoons \text{Bu} \cdot \text{S}$ (adsorption, equilibrium)
2. $\text{Bu} \cdot \text{S} \rightarrow \text{MEK} \cdot \text{S} + \text{H}_2\text{S}$ (surface reaction, RATE-LIMITING)
3. $\text{MEK} \cdot \text{S} \rightleftharpoons \text{MEK} + \text{S}$ (desorption, fast)
4. $\text{H}_2\text{S} \rightleftharpoons \text{H}_2 + \text{S}$ (desorption, fast)

(c) Six Ideas from Preface Table P-4:

1. Use systematic algorithm for rate law derivation
2. Sketch/plot data to visualize trends
3. Check limiting cases (high/low pressure)

(d) Conversion Calculation:

Estimated $k = 0.3583 \text{ (mol/min/kg)/atm}$

Catalyst weight for 90% conversion: $W = 6.43 \text{ kg}$

(e) Critical Thinking Question:

Question: Is our rate law valid at all pressures?

Why this requires critical thinking:

- We assume linear dependence on P_{Bu} (weak adsorption)
- Data only spans $P_{Bu} = 0.1$ to 2 atm
- At high P_{Bu} , saturation effects may occur
- At low P_{Bu} , mass transfer may limit the rate
- Extrapolation beyond data range is risky
- Need to verify assumptions hold in new conditions

(f) Accounting for Pressure Drop: